

Kayden Knapik

✉ kaydenknapik@gmail.com | [in linkedin.com/in/KaydenKnapik](https://www.linkedin.com/in/KaydenKnapik) | [globe kaydenknapik.com](https://kaydenknapik.com)

SUMMARY

Robotics Engineer specializing in bipedal robotics, combining mechanical design, simulation, and reinforcement learning for dynamic locomotion. Experienced in taking legged robots from CAD and manufacturing through sim-to-real deployment, with a focus on expressive robotic movement.

EDUCATION

Technical University of Eindhoven Eindhoven, NL
MSc in Mechanical Engineering - Robotics 2025 – 2027 (Expected)
Coursework: Optimal Control & RL, ML for Systems, Multi-body Dynamics, Software Engineering for AI

Technical University of Eindhoven Eindhoven, NL
BSc in Mechanical Engineering 2022 – 2025
Thesis: Development of an RL-based Control Pipeline for Bipedal Locomotion (BDX-R) | Grade: 9/10

TECHNICAL EXPERIENCE

Frame by Frame Rome, Italy
Bipedal Robotic Character Design Intern September 2026 – December 2026

- Designing a physical bipedal character with creative leads and animators, balancing aesthetics and mechatronic feasibility
- Designing custom CNC-manufactured structural components and developing the robot's mechanical architecture
- Developing actuator models, simulation, and RL pipelines for expressive locomotion and sim-to-real deployment

Tech United Student Team (TU Eindhoven) Eindhoven, Netherlands
Robotics Developer & Researcher 2023 – Present

- RoboCup MSL World Champion in 2024 and 2023 autonomous robot soccer tournaments
- Developing locomotion and dynamic skills for the T1 Humanoid in mjlab and contributing to sim-to-real deployment

ETH Zurich Robotics Club Hackathon Zurich, Switzerland
Robotics Developer (1st Place Winner) 2026

- Won 1st Place (\$5,000) by building a vision-based navigation and manipulation pipeline in 48 hours
- Integrated NVIDIA GR00T and LeRobot imitation learning for whole-body pick-and-place

KEY PROJECTS

Stompy: Rapid-Build 6-DOF Bipedal Robot | mjlab, RL, CAD, Jetson Orin Nano August 2026

- Designed and built a 6-DOF biped from scratch in under 72 hours for approximately \$1,300
- Built a fully 3D-printed chassis with NVIDIA Jetson Orin Nano onboard compute
- Trained and deployed an RL locomotion policy in mjlab for sim-to-real testing

BDX-R: Open Source Bipedal Robot | mjlab, Isaac Lab, CAD, PyTorch, ONNX 2025 – Present

- Engineered a 10-DOF biped with custom CAD for 3D printing, CNC, and sheet-metal manufacturing
- Built a PPO-based sim-to-real locomotion pipeline in Isaac Lab and mjlab
- Open-sourced the mechanical CAD and software stack for legged robotics research

DIY Robotics (YouTube Channel) | Robot Design, Prototyping, RL 2025 – Present

- Produce technical build logs on bipedal robotics, actuators, prototyping, and RL deployment

SKILLS

Simulation & RL : mjlab, Isaac Lab, IsaacSim, MuJoCo, PyTorch, ONNX
Programming : Python, C++, C, MATLAB
Hardware & Mechatronics : CAD (Fusion 360), CNC Design, URDF, 3D Printing, Soldering
Tools : Linux, Git, ROS2